JNJ-69086420 (JNJ-6420) (Johnson & Johnson (Janssen))

anti-Human kallikrein-2 monoclonal antibody for metastatic castration-resistant prostate cancer (mcrpc)

June 2026

IMPORTANT NOTICE

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^{225}Ac anti-hK2 TAT; Phase 1 only; RP2D undefined; rNPV $101M–$204M at 5% POS

Asset: JNJ-69086420 (JNJ-6420 — ^{225}Ac-DOTA-h11B6 anti-hK2 IgG1)

Stage: Phase 1 — NCT04644770 (primary completion December 2026)

Target: hK2 / KLK2 — AR-driven; prostate-restricted; first-in-class radioligand

Sponsor: J&J / Janssen

Asset

– Modality: targeted alpha therapy (antibody-based; internalizing)
– Indication: mCRPC post-ARPI; dedicated post-Pluvicto cohort
– Phase 1 PSA50 ~44% in biomarker-unselected patients
– Radiographic ORR 18% (n=17 measurable)

Value & Feasibility

– rNPV ~$101M base / ~$204M bull at 5% POS
– Peak sales unadjusted $1.4B–$2.8B
– WAC modeled $300K–$430K/course vs. Pluvicto confirmed floor $307K/6-cycle

Feasibility

– IP: h11B6 sequences; effective exclusivity ~2043–2044 (BPCIA)
– PSMA-RLT VISION trial partially pre-negotiates Phase 3 design
– Theranostic imaging pair (^{89}Zr-h11B6) ready for patient selection

Risk

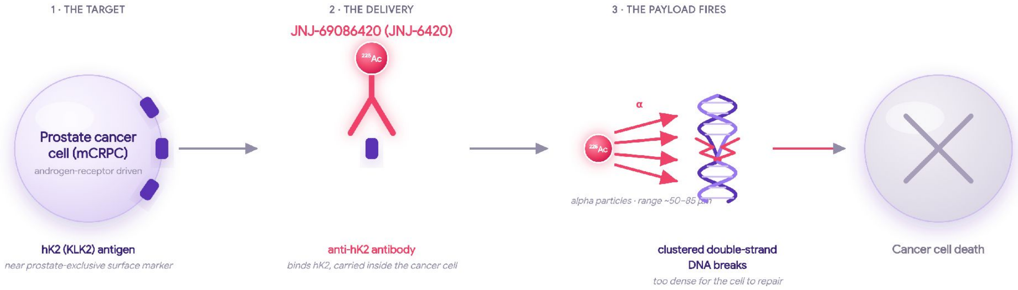
– Two fatal ILD events above 500 µCi; ^{225}Ac daughter nuclides redistribute by nuclear recoil; RP2D undefined
– ^{225}Ac global supply ~2,000 pts/year vs. ~3,900 required at base case
– Alpha-PSMA Phase 3 competitors 2–3 years ahead with confirmed RP2Ds

Valuation: M&A comps priced de-risked alpha-RLT at $1.4B–$2.4B; at 5% POS rNPV is $101M–$204M; December 2026 Phase 1 data adjudicates the RP2D gate

Sources: NCT04644770 — Phase 1 JNJ-69086420 (ClinicalTrials.gov) • ASCO 2024 Phase 1 results — JCO abstract (PSA50; ILD events) • PLUVICTO WAC $51,168/infusion — Buy & Bill A9607 (01/07/2026)

Castration resistance switches the hK2 beacon ON exactly as PSMA fades; JNJ-69086420 (JNJ-6420) homes to hK2 and delivers a short-range alpha payload the tumor can't repair

In castration-resistant prostate cancer, reactivated androgen-receptor (AR) signaling switches the hK2 (KLK2) protein ON at the cancer-cell surface — a precise homing beacon.



STANDARD RADIOLIGAND TODAY

PSMA-targeted beta emitter (^{177}Lu)

- Aims at PSMA — but reactivated AR signaling turns PSMA DOWN, so the homing beacon fades as the disease worsens.
- Beta particles travel far and hit softly — sparse damage, much of it single-strand and repairable by the tumor.
- Kidney / salivary-gland uptake limits the deliverable dose.

WHAT JNJ-69086420 (JNJ-6420) DOES DIFFERENTLY

hK2-targeted alpha emitter (^{225}Ac)

- Aims at hK2 (KLK2) — which AR signaling turns UP exactly when PSMA falls, so the beacon brightens as disease worsens.
- Alpha particles travel a few cell-widths and hit hard — dense, clustered double-strand breaks the tumor cannot fix.
- First-in-class targeted alpha therapy; Phase 1 in mCRPC.

Unresolved ILD and confirmed Phase 3 competition dominate the near-term risk profile

	Assessment	Observations	Rating
Clinical Value	**hK2 / KLK2 target biology**	– Rises as PSMA falls under AR signaling — mechanistically orthogonal to Pluvicto's niche	● ○ ○
	Phase 1 PSA50 efficacy signal	– ~44% in biomarker-unselected heavily pretreated patients on abstract-level data only	○ ● ○
	Safety / RP2D	– Two fatal ILD events above 500 µCi with no published mechanistic model and no RP2D defined	○ ○ ●
	Radiographic ORR	– 18% in only 17 measurable patients — below PoC threshold for a pivotal decision	○ ● ○
Commercial & Development	**IP position**	– h11B6 sequence patents with effective exclusivity to ~2043–2044 via BPCIA	● ○ ○
	Alpha-PSMA Phase 3 competition	– PSMAcTION and AlphaBreak have confirmed RP2Ds and are enrolling Phase 3 in the identical post-Pluvicto slot	○ ○ ●
	^{225}Ac supply constraint	– Global capacity ~2,000 patients/year versus ~3,900 required at the modeled base-case volume	○ ○ ●
	J&J internal prioritization	– Absent from ASCO 2025 after four-death disclosure while three competing prostate assets absorb capital	○ ○ ●

● No major issues ● Potential risks, addressable ● Major issues / no mitigation

Sources: NCT04644770 — Phase 1 JNJ-69086420 • NCT06780670 — PSMAcTION Phase 3 (Novartis, ^{225}Ac-PSMA-617) • NCT06402331 — AlphaBreak Phase 3 (AZ/Fusion, FPI-2265) • Wong et al. Clinical trial success rates. Biostatistics 2019

Alpha-PSMA agents reach Phase 3 with confirmed RP2D before JNJ-6420 completes Phase 1

Sources: NCT06780670 — PSMAcTION Phase 3 (Novartis) • NCT06402331 — AlphaBreak Phase 3 (AZ/Fusion FPI-2265) • FDA Pluvicto pre-taxane mCRPC label expansion • Novartis Q4 2025 — Pluvicto $2.0B FY2025

Abbreviations used in this assessment

Valuation & IP

POS probability of success
PTRS probability of technical & regulatory success

LOA likelihood of approval
rNPV risk-adjusted net present value

NPV net present value
DCF discounted cash flow
EV enterprise value
WAC wholesale acquisition cost
CoM composition-of-matter patent
PTE patent term extension
SPC supplementary protection certificate

BPCIA Biologics Price Competition & Innovation Act

COGS cost of goods sold

Mechanism & nonclinical

mAb monoclonal antibody
CTLA-4 cytotoxic T-lymphocyte antigen-4

ADC antibody-drug conjugate

Disease & clinical

OS overall survival
ORR objective response rate

Regulatory & trials

FDA US Food and Drug Administration

BLA Biologics License Application

Phase 1 hK2 ^{225}Ac-TAT in mCRPC; PSA50 44% but RP2D and ILD unresolved

Stage: Phase 1	Sponsor: Johnson & Johnson	Modality: ^{225}Ac targeted alpha therapy	Target: hK2 (KLK2) — PSMA-orthogonal prostate antigen

Asset

– hK2 rises as PSMA falls under AR reactivation, filling the post-Pluvicto/PSMA-low niche where no approved targeted RLT exists
– Alpha payload sidesteps ^{225}Ac-PSMA salivary-gland toxicity
– Combination pipeline with KLK2×CD3 TCE (pasritamig) and mHSPC line expansion

Value & Feasibility

– rNPV ~$75M–$250M (PTRS-composed POS 3–9%)
– Global peak revenue $1.4B–$2.8B risk-unadjusted
– Net price ~$200K–$230K/patient on Medicare Part B buy-and-bill

Feasibility

– PSA50 44% at 250 µCi (n=36, biomarker-unselected, heavily pretreated) clears empirical 40% PoC bar
– RECIST ORR 18% (17 evaluable measurable patients)
– Phase 1 primary completion Dec 2026

Risk

– 2 treatment-related ILD deaths at >500 µCi cumulative; daughter nuclide redistribution as a plausible unresolved pulmonary mechanism
– RP2D not defined
– Novartis and AZ/Fusion enrolling Phase 3 in the identical post-Pluvicto slot
– No ASCO 2025 presentation signals internal deprioritization

Valuation: rNPV $75M–$250M today; Dec 2026 Ph1 read on ILD resolution and RP2D definition is the primary re-rate catalyst

Sources: ASCO 2024 Phase 1 JNJ-6420 (JCO abstract) • NCT04644770 — Phase 1 ^{225}Ac-JNJ-69086420 • Pluvicto WAC $51,168/infusion — Jan 2026 • Novartis Q4 2025 — Pluvicto $2.0B FY2025

PSA50 44% clears PoC bar; composite POS 3–9% reflects ILD and modality risk

	Assessment	Observations	Rating
Efficacy Signal	**Phase 1 PSA50 (250 µCi)**	– 44% confirmed in 36 biomarker-unselected pretreated patients, clearing the empirical >40% PoC threshold	
	Radiographic ORR	– RECIST 18% in only 17 evaluable measurable patients, a limited denominator that is not confirmatory	
Safety	**ILD deaths**	– 2 treatment-related deaths at cumulative doses >500 µCi, with daughter nuclide redistribution as a plausible unresolved mechanism	
	RP2D status	– Not defined, with adaptive dosing ongoing and Phase 1 primary completion in Dec 2026	
Probability of Success	**Gosset PTRS (Ph1→Ph2)**	– 23.5%, with J&J sponsor track record and mCRPC disease history as primary uplifts	
	External LOA (Ph2→approval)	– 12–38% range (base 22%) from the oncology solid-tumor benchmark	
	Composite POS	– Point estimate 5.2% (range 3–9%), consistent with historical Ph1-to-approval base rates in oncology	

No major issues

Potential risks, addressable

Major issues / no mitigation

Sources: ASCO 2024 Phase 1 JNJ-6420 (JCO abstract) • NCT04644770 — Phase 1 ^{225}Ac-JNJ-69086420 • Wong et al. 2019 — Ph2→approval oncology LOA • Hay et al. 2014 — Ph1→approval oncology base rate

rNPV $75M–$250M on PTRS-composed POS; Dec 2026 Phase 1 read is the adjudicating catalyst

Price anchor: ~$200K–$230K net/patient on Pluvicto-anchored WAC via Medicare Part B (gross-to-net ~15%)

Addressable pool: 50K incident US mCRPC × 65% systemically fit × 60% RLT-eligible × 20–35% peak share = 3,900–6,825 patients

Scenario	Global peak revenue	POS (PTRS-composed)	rNPV
Bear	$1.4B	3%	~$40M
Base	$1.4B	5%	~$75M
Bull	$2.8B	9%	~$250M

Sources: Pluvicto WAC $51,168/infusion — Jan 2026 • CADTH pharmacoeconomic review — Pluvicto cost-effectiveness • ASCO 2024 Phase 1 JNJ-6420 (JCO abstract) • Wong et al. 2019 — oncology LOA benchmark

PSMA-low and post-Pluvicto mCRPC has no approved targeted radiotherapy

mCRPC treatment is sequenced by prior exposure; every approved line carries a structural ceiling.

PSMA-low patients (~20–40% of mCRPC) are ineligible for all approved PSMA-targeted agents

Non-BRCA HRR: PARP rPFS benefit is narrow (HR ~0.74 vs ~0.20 in BRCAm), limiting combinatorial access

Line	Agent	Mechanism	Efficacy landmark	Key ceiling
1L (ARPI-naïve)	Abiraterone or enzalutamide	CYP17i / ARSi	OS HR 0.74–0.81	ARPI cross-resistance forecloses sequential re-use
1L (BRCAm)	Olaparib+abi / niraparib+abi	PARP + CYP17i	rPFS HR 0.24–0.53	Benefit attenuates sharply in non-BRCA HRR (HR ~0.74)
Post-ARPI	Docetaxel / cabazitaxel	Taxane	OS HR 0.70–0.76	Severe myelosuppression; functional reserve required
Post-ARPI/taxane (PSMA+)	Pluvicto (177Lu-PSMA-617)	PSMA beta-RLT	OS HR 0.62; rPFS HR 0.41	PSMA-PET required; PSMA-low patients excluded
Post-Pluvicto / PSMA-low	No approved targeted option	—	—	Cabazitaxel is the de facto fallback

Sources: NCCN — Prostate Cancer Guidelines • Sartor et al. VISION trial — 177Lu-PSMA-617. *NEJM* 2021 • FDA — Pluvicto pre-taxane mCRPC label expansion

Phase 1 PSA50 44% clears the pre-PoC bar; ILD deaths and undefined RP2D remain open

	Assessment	Observations	Rating
JNJ-6420 (225Ac-hK2 TAT) — Phase 1 Evidence Assessment (NCT04644770)	**PSA50 response rate**	– ~44% in heavily pretreated, biomarker-unselected mCRPC	● ○ ○ (green)
	Radiographic ORR	– 18% confirmed ORR in only 17 measurable patients, abstract-level data, no CIs disclosed	○ ● ○ (orange)
	Target differentiation	– hK2 rises as PSMA falls under AR reactivation, mechanistically orthogonal to all approved PSMA agents	● ○ ○ (green)
	ILD/pulmonary toxicity	– 2 ILD deaths above 500 µCi cumulative dose, 4 total TRDs presented at ASCO 2024, mechanism unpublished	○ ○ ● (red)
	RP2D and timeline	– Recommended Phase 2 dose not yet defined, adaptive dose-capping ongoing, primary completion December 2026	○ ○ ● (red)
	Sponsor prioritization	– Absent from ASCO 2025 agenda, three competing J&J prostate assets absorbing portfolio capital	○ ○ ● (red)

● No major issues ● Potential risks, addressable ● Major issues / no mitigation

Sources: ASCO 2024 — JNJ-6420 Phase 1 abstract (JCO) • NCT04644770 — Phase 1 225Ac-JNJ-69086420 • Sartor et al. VISION — PSMA RLT regulatory template. *NEJM* 2021

rNPV $101M–$204M at 5% POS; Phase 2 data resolving ILD are the sole re-rating catalyst

US patient funnel (steady-state annual)

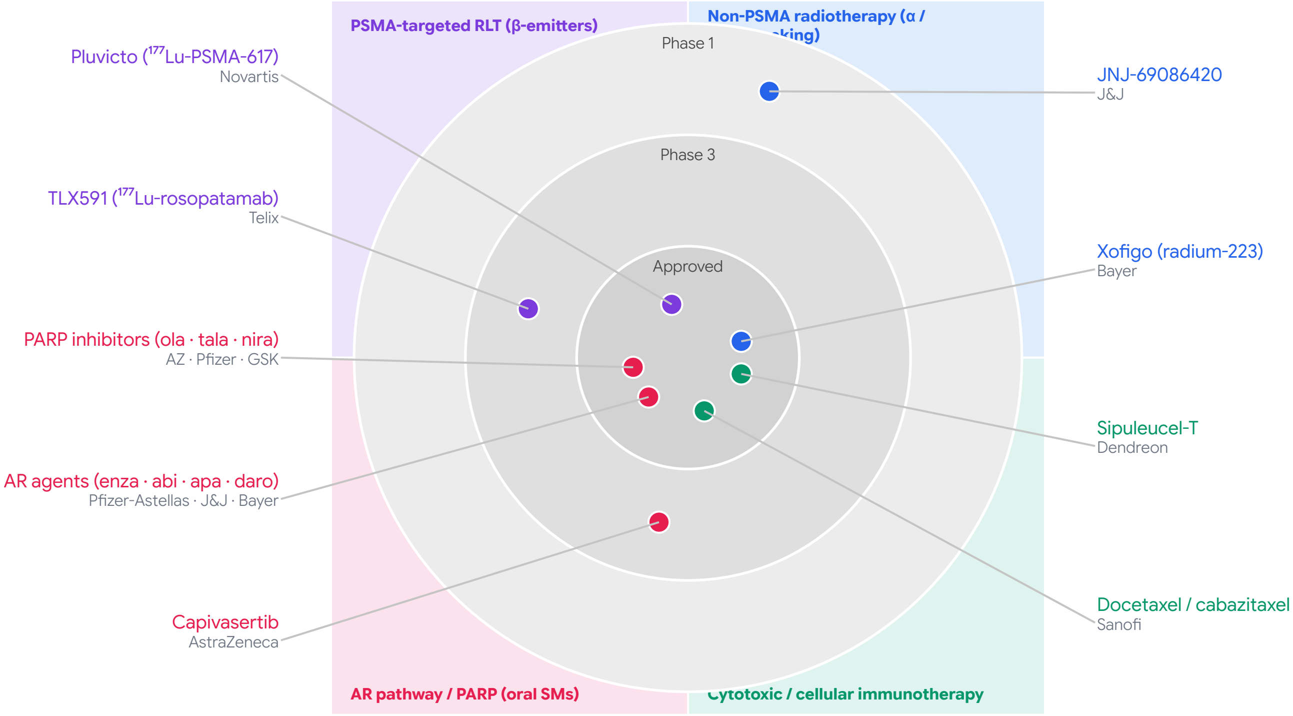
~50,000 incident mCRPC → 32,500 post-ARPI/ECOG 0–1 → 19,500 RLT-addressable → **3,900–6,825** peak-treated (20–35% share, ~$200–230K/yr net)

POS = Ph1→Ph2 23.5% × Ph2→approval ~22%, discounted for alpha-TAT novelty and open ILD signal. Global 1.8× uplift yields $1.4B–$2.8B worldwide peak; figures exclude COGS and full PV haircut.

Scenario	Net peak revenue	NPV @ approval (~2.5×)	POS	rNPV
Base	$780M	$1.95B	5%	**$101M**
Bull	$1.57B	$3.92B	5%	**$204M**

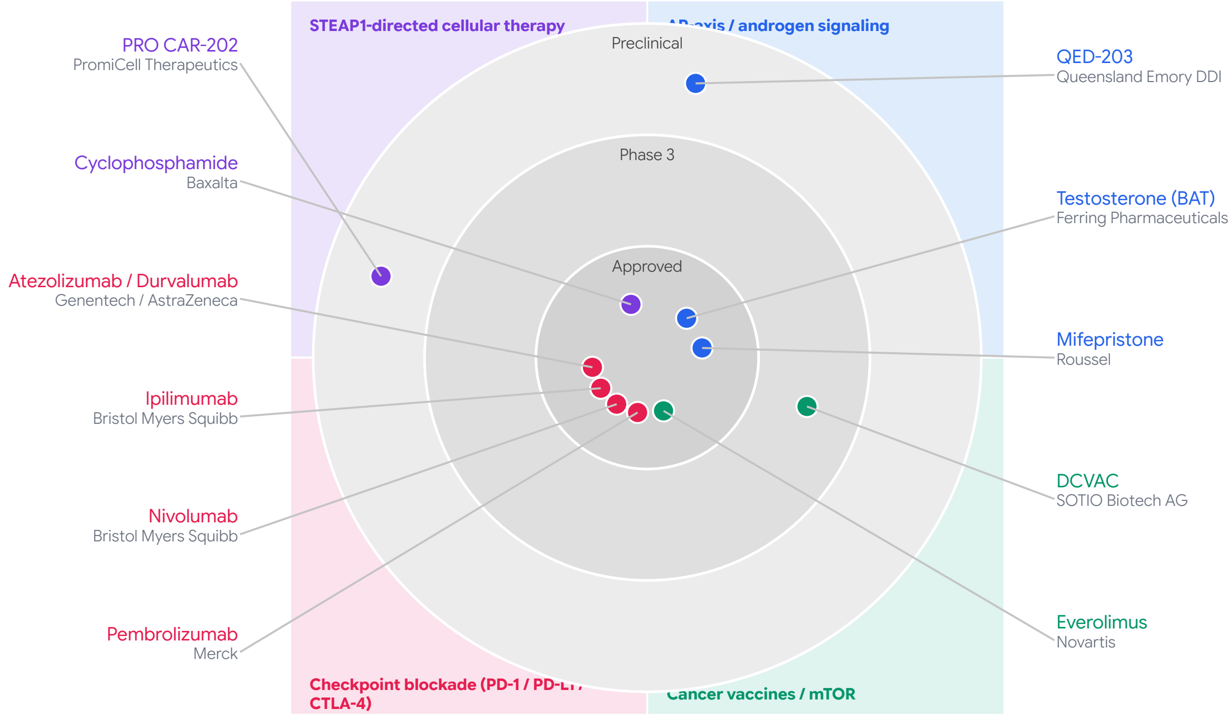
Sources: Wong CH et al. Clinical trial success rates. *Biostatistics* 2019 • Hay M et al. Clinical development success rates. *Nat Biotechnol* 2014 • PLUVICTO WAC $51,168/infusion — Buy & Bill A9607 • Novartis Q4 2025 Results — Pluvicto $2.0B FY2025

hK2 gives JNJ-6420 the only non-PSMA tumor-antigen RLT position in mCRPC



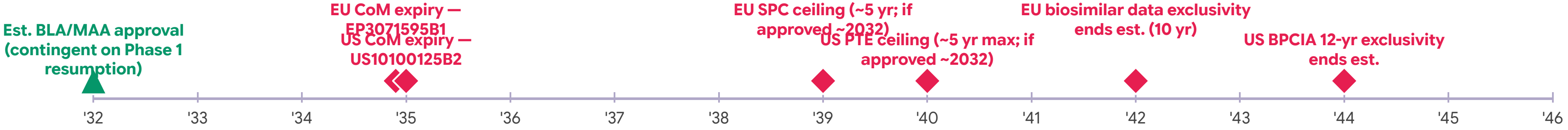
Sources: ASCO 2024 — Phase 1 JNJ-6420 (PSA50 ~44%, ILD DLTs) • Sartor O et al. VISION trial — ^{177}Lu-PSMA-617 in mCRPC, NEJM 2021 • FDA Pluvicto pre-taxane mCRPC label expansion • NCT04644770 — Phase 1 ^{225}Ac-JNJ-69086420

PD-1/CTLA-4 blockade dominates early mCRPC; STEAP1 cellular space thin



Sources: [NCT04644770 — Phase 1 ^{225}Ac-JNJ-69086420 (hK2 α-RLT)] • [ASCO 2024 — Phase 1 JNJ-6420 abstract]

US/EU core CoM patents to 2034–2035; PTE/SPC could extend to ~2040



Sources: US10100125B2 — h11B6 CoM patent; Janssen Biotech; exp. Jan 2035 • EP3071595B1 — EU CoM counterpart; exp. Nov 2034 • US12077585B2 — Next-gen KLK2 ABD blocking patent; exp. Oct 2040 • WO2013061083A1 — Origination murine 11B6 patent; Fredax; ~Oct 2032

hK2 restricts expansion to prostate; non-prostate data are expression-only

All four active NCT04644770 cohorts are prostate-cancer only; earlier lines (biochemical recurrence, nmCRPC) are mechanistically plausible but unenrolled.

Take-home: The expansion space is real but prostate-bounded; theranostic ^{111}In/^{89}Zr-DOTA-h11B6 imaging is the only non-trial evidence of in-vivo target engagement and the primary case for extending into earlier lines.

Candidate indication	Evidence tier	In trial?
mHSPC (hormone-sensitive, earlier state)	Clinical — protocol cohort	Yes — NCT04644770 Part 4a
Oligometastatic PCa + SBRT combination	Clinical — protocol cohort	Yes — NCT04644770 Part 4b
Combination with pasritamig (hK2×CD3 TCE)	Clinical — protocol cohort	Yes — NCT04644770 Part 3
Post-^{177}Lu-PSMA mCRPC	Clinical — protocol cohort	Yes — NCT04644770 Part 2c
Biochemical recurrence / nmCRPC / neoadjuvant	Mechanistic + theranostic imaging	No — plausible future vector
Breast cancer	Expression only — no antigen-density POC	No
Salivary / thyroid / other solid tumors	Negligible evidence	No

Sources: NCT04644770 — Phase 1 JNJ-69086420 (ClinicalTrials.gov) • ASCO 2024 Phase 1 JNJ-6420 results — JCO abstract

Licensing median \$50M upfront / \$572M total; ^{225}Ac M&A median ~\$2.2B

14 benchmarked deals; licensing and M&A set distinct value anchors for a pre-RP2D asset.

Take-home: Licensing median \$50M upfront / \$572M total; ^{225}Ac-prostate M&A \$1.4–2.9B (median ~\$2.2B); FPI-2265 at \$2.4B post-RP2D — a bar JNJ-6420 has not yet cleared.

Date	Parties	Asset	Type	Upfront	Total
2020-01	Diaprost → top-10	h11B6 (anti-hK2; same target)	Option→License	\$13M	~\$103M
2017-11	Zymeworks → Janssen	pasritamig (KLK2×CD3 TCE)	License	\$50M	~\$1,452M
2025-06	Philochem → BMS/RayzeBio	OncoACP3 (^{225}Ac radioligand)	License	\$350M	~\$1,350M
2024-07	Full-Life → SK Biopharma	FL-091 (NTSR1 ^{225}Ac)	License/option	n/d	\$571.5M
2018-10	Endocyte → Novartis	^{177}Lu-PSMA-617 (Pluvicto)	M&A	n/a	\$2,100M
2024-03	Fusion → AstraZeneca	FPI-2265 (^{225}Ac-PSMA; closest comp)	M&A	\$2,000M	\$2,400M
2024-05	Mariana → Novartis	^{225}Ac platform (prostate)	M&A	\$1,000M	~\$1,750M
2014-01	Algeta → Bayer	Xofigo (^{223}Ra; first alpha-emitter)	M&A	n/a	\$2,900M

Sources: Fusion / AstraZeneca acquisition (AstraZeneca IR) • Endocyte + Mariana / Novartis acquisitions (Novartis IR) • Algeta / Bayer acquisition (Bayer IR) • Philochem / RayzeBio deal (Bristol Myers Squibb IR)

Summary

Gate-dependent rNPV — ILD mechanism and competitor pace are binary adjudicators

	Assessment	Observations	Rating
Commercial Logic (Real)	**PSMA-orthogonal antigen**	– hK2 rises as PSMA falls under AR reactivation — non-overlapping with PSMA-RLT patient selection	● ○ ○
	Post-Pluvicto slot	– No approved targeted radiotherapy in PSMA-low / post-PSMA-617 disease	● ○ ○
	Alpha payload differentiation	– Antibody vector avoids ^{225}Ac-PSMA xerostomia – KLK2×CD3 doublet with pasritamig in Phase 3 planning	○ ● ○
Investment Case (Pre-PoC)	**ILD / treatment-related deaths**	– Two TRDs at >500 µCi – ^{225}Ac daughter redistribution is a potential modality-level ceiling	○ ○ ●
	RP2D undefined	– No declared safe dose – radiographic ORR 18% in n=17 measurable patients	○ ○ ●
	Competitor phase gap	– PSMAcTION (Novartis) and AlphaBreak (AZ/Fusion) enrolling Phase 3 with confirmed RP2Ds and contracted isotope supply	○ ○ ●
	J&J portfolio signal	– No ASCO 2025 presentation – three competing prostate assets absorbing capital in same indication	○ ● ○

● No major issues ● Potential risks, addressable ● Major issues / no mitigation

Sources: NCT04644770 — Phase 1 ^{225}Ac-JNJ-69086420 • ASCO 2024 Phase 1 JNJ-6420 results • PSMAcTION — NCT06780670 • AlphaBreak — NCT06402331

Four binary catalysts adjudicate whether rNPV enters the alpha-RLT deal range

ILD mechanism classification (binary gate): J&J case narratives + ^{225}Ac daughter dosimetry determine whether fatal pneumonitis is a modality ceiling from daughter redistribution or a manageable therapeutic-window problem — adverse resolution makes remaining catalysts moot

NCT04644770 primary readout (Dec 2026): RP2D declaration ≤500 µCi with ILD incidence near zero AND confirmed RECIST response in Part 2c post-Pluvicto cohort — partial signal on either axis does not de-risk value

Competitor Phase 3 pace: Positive PSMAcTION or AlphaBreak pivotal readout before JNJ-6420 Phase 3 initiation closes the post-Pluvicto slot to a second alpha entry — timing risk is acute given a December 2026 Phase 1 primary completion

KLK2/PSMA co-expression in post-Pluvicto tissue: Dual biomarker analysis on archived biopsy confirms hK2 retention in AR-bypass / lineage-plasticity-enriched tumors at RLT progression — if negative, the core PSMA-low targeting thesis fails regardless of safety outcome

Take-home: All four must resolve favorably and in sequence to support rNPV re-rating toward the $1.4B–$2.9B alpha-RLT comparable range; ILD mechanism is the first and highest-leverage gate.

Sources: NCT04644770 — Phase 1 ^{225}Ac-JNJ-69086420 • PSMAcTION — NCT06780670 • AlphaBreak — NCT06402331 • ASCO 2024 Phase 1 JNJ-6420 results

Key sources for this assessment

Asset Overview — NCT04644770 — Phase 1 JNJ-69086420 (ClinicalTrials.gov) · ASCO 2024 Phase 1 results — JCO abstract (PSA50; ILD events) · PLUVICTO WAC $51,168/infusion — Buy & Bill A9607 (01/07/2026) · NCT06780670 — PSMAcTION Phase 3 (Novartis, ^{225}Ac-PSMA-617) · NCT06402331 — AlphaBreak Phase 3 (AZ/Fusion, FPI-2265) · Wong et al. Clinical trial success rates. Biostatistics 2019 · FDA Pluvicto pre-taxane mCRPC label expansion · Novartis Q4 2025 — Pluvicto $2.0B FY2025 · Hay et al. 2014 — Ph1→approval oncology base rate · CADTH pharmacoeconomic review — Pluvicto cost-effectiveness

Disease Overview — NCCN — Prostate Cancer Guidelines · Sartor et al. VISION trial — 177Lu-PSMA-617. *NEJM* 2021

Competitive Landscape — US10100125B2 — h11B6 CoM patent; Janssen Biotech; exp. Jan 2035 · EP3071595B1 — EU CoM counterpart; exp. Nov 2034 · US12077585B2 — Next-gen KLK2 ABD blocking patent; exp. Oct 2040 · WO2013061083A1 — Origination murine 11B6 patent; Fredax; ~Oct 2032

Indication Expansion — Fusion / AstraZeneca acquisition (AstraZeneca IR) · Endocyte + Mariana / Novartis acquisitions (Novartis IR) · Algeta / Bayer acquisition (Bayer IR) · Philochem / RayzeBio deal (Bristol Myers Squibb IR)

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